

KEERTHAN MAHESH

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WORK EXPERIENCE

Wells Fargo

Senior Software Engineer

Charlotte, NC

Jun 2026 - Present

- Spearheaded the engineering of core **Risk & Governance platforms**, drove the architecture for building scalable systems to manage complex corporate risk hierarchies, major requirements planning and scoping across Risk/Control Assessable Units.
- **Architected database schemas** and managed **database migrations** using **Liquibase DDL/DML scripts**, creating **strategic indexes**, ensuring seamless data model evolution and data integrity across environments.
- **Automated CI/CD deployments and observability**, orchestrating containerized application delivery via **Harness** to **OpenShift Container Platform (OCP)** while leveraging **Splunk** for centralized logging, real-time alerting, and rapid issue resolution.

KonfigAI

Backend Lead Engineer

Charlotte, NC

May 2023 - May 2026

- **Sole architect of the main microservice** for **KonfigAI**, a no-code platform, responsible for the architecture and design of a production-level web application; developed **Agentic AI** capabilities enabling users to generate entire applications from a single prompt; a **multi-agent system** pairing an **LLM orchestrator** with **18 specialized sub-agents** for workflow automation.
- Designed the companion **MCP tool server** (FastMCP over SSE) exposing 100+ Konfig operations; with **tenacity-based retries**, **async connection pooling**, per-request header forwarding, and **token-optimized response encoding**; integrated **7 LLM providers**.
- Optimized **JVM performance** using **GC (G1) tuning**, **-Xms/-Xmx heap sizing**, **pause-time targets**, **heap dump analysis**, and **Java 21 features (Records, Virtual Threads)**, reducing application memory limits by 60%; increasing concurrent users by 200%.
- Built a standalone **AI agent** to **review merge requests**, **suggest enhancements**, **remediate SonarQube issues**, **resolve build failures**, and **analyze Trivy vulnerability scan results**, improving code quality and security review efficiency.
- Developed high-performance, responsive UI using **React.js** and **React Redux**, implementing a **component-driven architecture** using **custom hooks** and efficient **state management**. Optimized application delivery through **code-splitting** and **lazy loading**.

Spatial Modeling and Remote Sensing Lab, University of Maryland

Research Assistant - Advisor: Dr. Varaprasad Bandaru - SMarS Lab

College Park, MD

Aug 2022 - May 2023

- **Nitrogen Recommendation Tool**: Built an end-to-end **React + Node.js** decision support web application featuring **React Leaflet** **geospatial mapping** that optimized location-based data processing for over **100+ farmers across Oklahoma and Kansas**.

KonfigAI

Software Engineer

Hyderabad, India

Aug 2020 - Aug 2022

- Implemented and optimized scheduled batch operations using **Spring Batch**; processed **200,000 complex transactions** in under **5 minutes**, significantly improving system performance and efficiency.
- Designed a multi-layer **caching strategy** using **Redis (LRU cache eviction strategy, custom TTL)** for **distributed caching** and **Caffeine** for in-memory caching for **connection pool managers** across **MongoDB, MySQL, PostgreSQL, DynamoDB, and SAP HANA**, improving latency, throughput, and backend resource utilization.
- Implemented secure authentication and authorization using **Okta** and **Keycloak**, **JWT token validation**, custom **OAuth 2.0 grant flows - Authorization Code, Implicit, Client Credentials, ROPC - OIDC integration**, **OWASP Top 10 controls** and **role-based access control (RBAC)** to enhance security for user applications.
- Collaborated with **DevSecOps** team using **Jenkins** for **CI/CD automation**, **Helm charts** for **Kubernetes deployments**, **Rancher** for **container orchestration**, **Grafana** for **log monitoring**, **GC monitoring**, **CPU utilization dashboards**, **OpenTelemetry** for **observability**, and **OpenBao** for **secrets management**, improving deployment reliability and operational visibility.
- Used **JProfiler** and **Cryostat** for **JVM profiling**, **runtime diagnostics**, **memory leak analysis**, and **performance monitoring**.
- Engineered comprehensive testing strategy - **unit tests** using **JUnit** and **Mockito**, **embedded database tests**, **integration tests**, and **load testing** using **k6** to validate application reliability, API behavior, and performance under high traffic.

JP Morgan Chase & Co.

Software Development Intern

Hyderabad, India

Jul 2019 - Dec 2019

- Engineered a **React + Golang** web application that visualizes **Kubernetes entity relationships** using **D3 graph**.

PROJECTS

Resume Intelligence: Retrieval-Augmented Generation (RAG) app for Job-Description ↔ Resume matching

- An end-to-end **RAG pipeline**: document ingestion (PDF/DOCX parsing) → **semantic chunking** → **embedding generation** (text-embedding-3-large) → **HNSW-based Approximate Nearest Neighbors (ANN) retrieval** over a **vector store** (Weaviate) → **cross-encoder reranking** → **LLM-driven scoring**, gap analysis, and tailoring recommendations using **FastAPI**.

EDUCATION

University of Maryland, College Park (M.Eng., Software Engineering)

Aug 2022 - Dec 2023

Birla Institute of Technology and Science, Pilani (B.Eng., Electrical and Electronics Engineering)

Aug 2016 - Aug 2020

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python, Go, Bash, Assembly, SQL, HTML, CSS, JavaScript, TypeScript, JSON, XML.

Tools and Frameworks: LangGraph, LangChain, MCP, FastAPI, React, React Native, Angular, Spring Boot, Gradle, MongoDB, MySQL, AWS, Kafka, CI/CD, SonarQube, Kubernetes, Docker, Grafana, Jenkins, Spring IoC, Spring JPA, Spring AOP, REST, SOAP.

PUBLICATIONS AND TEACHING

- Agarwal S., Mahesh K., Singh P., Shah J., Sanjeev N. **Investigating Depression Semantics on Reddit using Deep Learning**.
- Undergraduate Teaching Assistant for **Object-Oriented Programming in Java - CS F213**.